

MONTEGO BAY SANGSTER, Jamaica

WMO: 783880

Lat: 18.504N

Lon: 77.913W

Elev: 4

StdP: 14.69

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	71.5	72.9	60.7	79.3	75.4	62.7	85.2	76.0	25.4	79.8	23.2	79.9	4.4	140	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	91.4	81.0	90.0	80.6	89.6	80.5	83.6	88.7	82.7	87.6	82.0	87.0	16.5	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.3	168.5	87.6	81.0	161.1	86.4	80.6	159.1	86.1	47.7	88.4	46.4	88.3	45.8	87.4	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.7	20.6	18.7	DB	69.3	94.0	2.1	1.0	67.7	94.8	66.5	95.3	65.3	95.9	63.7	96.6
			WB	63.4	83.9	2.2	1.5	61.8	85.0	60.5	85.8	59.3	86.6	57.7	87.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.5	78.4	78.6	79.1	80.7	81.9	83.5	84.0	84.1	83.6	82.6	81.4	79.7	
	DBStd	2.62	1.76	1.55	1.80	1.42	1.61	1.61	1.40	1.51	1.59	1.67	1.74	1.84	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD50	11496	882	801	903	920	989	1005	1055	1058	1009	1011	941	922	
	CDD65	6021	417	381	438	470	524	555	590	593	559	546	491	457	
	CDH74	63278	3249	3027	3744	4683	5650	6669	7254	7207	6566	6073	5061	4095	
	CDH80	20202	522	578	786	1224	1780	2617	2974	2946	2504	2039	1382	850	
Wind	WSAvg	7.7	8.7	8.4	8.4	7.8	7.1	7.7	8.1	7.2	6.0	6.3	7.9	8.5	
Precipitation	PrecAvg	44.7	2.4	1.8	2.2	2.6	5.2	4.2	2.6	4.0	5.8	6.7	4.6	4.2	
	PrecMax	65.1	9.2	7.9	6.5	12.8	17.6	13.0	11.7	12.0	16.7	26.5	14.1	15.0	
	PrecMin	24.4	0.4	0.0	0.0	0.0	0.0	0.2	0.6	0.9	1.2	0.9	0.6	0.6	
	PrecStd	11.3	2.0	1.8	1.9	2.8	3.7	2.9	2.2	2.5	3.1	4.9	3.1	3.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.2	86.4	87.8	88.2	89.8	92.0	92.1	92.6	91.8	91.1	89.6	87.9	
		MCWB	75.8	75.8	77.2	78.0	79.6	79.9	81.4	81.8	82.0	81.2	81.0	79.4	
	2%	DB	84.6	85.3	86.0	86.6	88.2	90.1	90.8	91.1	90.0	89.3	87.9	86.1	
		MCWB	76.1	75.6	76.7	77.7	79.5	80.0	80.7	81.5	81.0	80.6	80.2	77.9	
	5%	DB	84.1	84.3	84.6	86.0	87.6	89.6	89.7	89.8	89.4	88.0	86.4	84.7	
		MCWB	76.1	75.5	76.1	77.6	79.3	80.0	80.5	80.9	80.7	80.2	79.2	77.2	
	10%	DB	82.7	83.4	84.0	84.9	86.3	88.2	88.8	89.0	88.1	87.4	85.8	84.2	
		MCWB	75.4	75.1	75.8	76.9	78.6	79.8	80.2	80.6	80.3	79.9	78.7	76.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.9	78.9	80.0	81.4	83.3	83.4	84.4	84.5	84.3	83.9	83.1	82.1
			MCDB	82.5	83.0	83.2	84.9	86.1	88.6	90.1	90.3	89.9	88.7	86.5	84.9
2%		WB	78.6	77.8	78.8	80.0	81.7	82.3	82.7	83.0	82.8	82.2	81.8	79.7	
		MCDB	82.1	82.6	83.1	84.5	85.7	87.6	88.4	88.6	88.0	86.6	85.5	83.4	
5%		WB	77.4	76.9	77.8	79.1	80.5	81.3	81.8	82.2	82.0	81.2	80.6	78.8	
		MCDB	81.8	82.3	82.7	84.1	85.2	87.2	87.7	87.7	87.2	86.0	84.8	83.1	
10%		WB	76.5	76.1	76.9	78.1	79.7	80.4	80.9	81.2	81.1	80.4	79.7	77.9	
		MCDB	81.3	81.7	82.3	83.5	84.7	86.7	87.1	87.2	86.7	85.5	84.0	82.5	
Mean Daily Temperature Range	5% DB	MDBR	9.2	10.0	9.9	9.7	10.0	10.8	10.9	10.9	10.5	9.8	9.0	9.1	
		MCDBR	10.5	11.2	10.8	10.7	10.8	11.7	11.7	11.7	11.3	10.6	10.0	10.1	
		MCWBR	6.7	7.3	6.8	6.7	7.1	7.4	7.5	7.2	6.8	6.6	6.0	6.2	
	5% WB	MCDBR	9.4	10.4	10.2	9.9	10.2	10.8	10.9	10.9	10.5	9.8	9.6	9.3	
		MCWBR	6.6	7.3	7.0	6.7	7.3	7.6	7.7	7.4	7.2	6.6	6.3	6.2	
Clear-Sky Solar Irradiance	taub	0.380	0.382	0.388	0.420	0.447	0.491	0.500	0.462	0.423	0.394	0.387	0.378		
	taud	2.533	2.511	2.491	2.403	2.364	2.228	2.186	2.335	2.453	2.554	2.540	2.560		
	Ebn at Noon	287	292	292	282	271	257	255	266	278	284	282	284		
	Edh at Noon	31	33	35	39	40	45	47	41	36	32	31	30		
All-Sky Solar Radiation	RadAvg	1404	1650	1863	1992	1887	1868	1958	1897	1770	1541	1418	1335		
	RadStd	82	70	67	77	108	98	112	65	130	125	99	75		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page